CHAPTER

18

Legal Aspects of DNA Testing and the Scientific Expert in Court

INCLUDING INPUT FROM INTERVIEWS WITH DR. ROBIN COTTON, TED HUNT, BRAD BANNON, THE ORANGE COUNTY (CA) DISTRICT ATTORNEY'S DNA UNIT, AND DR. CHARLOTTE WORD

There are important differences between the quest for truth in the courtroom and the quest for truth in the laboratory. Scientific conclusions are subject to perpetual revision. Law, on the other hand, must resolve disputes finally and quickly (U.S. Supreme Court Justice Harry Blackmun in the 1993 decision of Daubert v. Merrell Dow Pharmaceuticals, Inc.*).*

As noted on the American Academy of Forensic Sciences website (http://www.aafs.org), forensic science is the application of science to the just resolution of legal issues. Thus, forensic DNA scientists produce results that directly impact the criminal justice system. These results and their subsequent conclusions are usually presented in the form of a case report. DNA analysts are sometimes called upon to testify in court regarding the conclusions derived from the test results produced during the course of their examination of the evidence provided to them and recorded in their case reports. The Southwestern Association of Forensic Scientists' Code of Professional Conduct reminds their members that. "Forensic scientists…have an obligation to provide opinions and facts to a court of law that are truthful, honorable, and of sound judgment… The work product of the forensic scientist is culminated in the conclusions stated in their report and during testimony." (see http://www.swafs.us/pdf/2009/2008bylaws.pdf). The purpose of this chapter is to discuss the roles and responsibilities of expert witnesses in the framework of the U.S. legal system to aid DNA analysts preparing to testify in court. Lawyers and judges need to understand some science

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to do their jobs well. Likewise, scientists should comprehend and appreciate some law to serve effectively in court.

Although I am not permitted to testify as an expert witness because of my current position as a federal government employee at the National Institute of Standards and Technology, I have been asked to do so many times. I have taken courtroom testimony training and given a lot of thought to the issues involved from my perspective on the sidelines of courtroom drama. Over the past few years, I have also had the privilege of talking to and teaching numerous lawyers (both prosecuting and defense attorneys) about DNA testing. Based in large part on these discussions, I feel a need to add this chapter to Advanced Topics in Forensic DNA Typing: Methodology *to help bridge the divide between science and law that can and does exist for many people. As a disclaimer, the material provided herein is not intended to be complete in its coverage of legal issues. Remember that I am a scientist and not a lawyer! I welcome feedback from members of the legal community and forensic DNA scientists in order to improve and expand the content in future editions.*

In addition to what I have gleaned from studying the resources listed at the back of this chapter, I have interviewed several individuals who have extensive experience as expert witnesses or other roles in the courtroom. In covering various topics, I will include direct quotes from these interviews (citing the individual contributions through their initials). Complete responses to interview questions are available in Appendix 4 at the back of this book. Prior to submission of this text, all those interviewed reviewed what I have written in order to confirm that I have quoted them correctly and placed their words in an appropriate context.

(RWC) Robin W. Cotton, Ph.D., is associate professor and director of the biomedical forensic sciences program at Boston University. She has over 20 years experience with forensic DNA analysis and served as the Laboratory Director for Cellmark Diagnostics from 1994 to 2006. Robin has served as an expert witness and testified in over 200 trials regarding DNA evidence including the O.J. Simpson trial in 1995. For more information, see http://www.bumc.bu.edu/biomedforensic/faculty-and-staff/faculty/robin-cotton/.

(CJW) Charlotte J. Word, Ph.D., is currently a private consultant with over 20 years of experience in forensic DNA analysis. She spent 15 years at Cellmark Diagnostics, in Germantown, Maryland, and was one of the Laboratory Directors there. She has testified in numerous admissibility hearings and in over 200 trials regarding DNA evidence. Charlotte also serves on the editorial board for the *Journal of Forensic Sciences* and was a member of the Postconviction Issues Working Group of the National Commission on the Future of DNA Evidence.

(TRH) Ted R. Hunt, J.D., is Chief Trial Attorney at the Jackson County Prosecutor's Office in Kansas City, Missouri. He has been a prosecuting attorney for 20 years and tried over 100 felony cases, most of which involved the presentation of DNA evidence. He is also actively involved in training attorneys and law enforcement officials in forensic DNA-related litigation for a number of state and federal organizations.

(OCDA) Orange County (California) District Attorney's DNA Unit is comprised of experienced deputy district attorneys who possess expertise regarding the utilization of forensic DNA evidence to solve crime. The OCDA DNA Unit provides assistance and training to all of the county's law enforcement agencies. Members of the OCDA DNA Unit, which include attorneys Bruce Moore, Camille Hill, Scott Scoville, Jennifer Contini, Terry Cleaveland, and Tammy Spurgeon, work closely with local law enforcement investigators, prosecutors, crime lab forensic scientists and OCDA local DNA database forensic scientists to effectively use forensic DNA technology and resources.

(BB) Bradley Bannon, J.D., is a criminal trial defense attorney for the law firm of Cheshire, Parker, Schneider, Bryan, and Vitale in Raleigh, North Carolina. He has been practicing law for 13 years and has helped draft legislation to strengthen the right of the accused to pretrial discovery in criminal cases in North Carolina. He was instrumental in using DNA evidence to help exonerate the Duke lacrosse players of false accusations of rape in 2006 and 2007. For more information, see http://www.cheshirepark.com/bbannon.html.

The 2001 Scientific Working Group on DNA Analysis Methods (SWGDAM) Training Guidelines emphasize in section 7 on legal issues that a DNA analyst trainee should be instructed on the legal system of his/her own jurisdiction (SWGDAM 2001). Furthermore, these guidelines suggest that a trainee should receive instruction on the following topics: (1) courtroom procedures and rules of evidence, (2) examiner/analyst qualifications, (3) technical testimony, (4) courtroom demeanor and attire, (5) testimony practice, (6) moot court(s), (7) discovery and admissibility rules, (8) ethical responsibility of an expert witness, (9) court system structure, and (10) evidence presentation. In addition, the examiner/analyst is encouraged to prepare a curriculum vitae and observe expert testimony, read relevant and appropriate transcripts or pertinent case law, and successfully complete at least one moot court (SWGDAM 2001). This chapter and the accompanying interview transcripts in Appendix 4 are intended to benefit this instruction for DNA analysts operating in U.S. courts of law.

SCIENCE AND LAW: DIFFERENT PERSPECTIVES

Several years ago I was asked by the American Prosecutor's Research Institute (APRI) to help prepare a DNA training course for lawyers. I and one other scientist were in a room with four prosecutors and an educational consultant for two days. I recall hearing terms like "Brady," "Frye," and "admissibility" being used regularly without any explanation. To these lawyers, a mere mention of "Brady" brought a concept of the requirement for disclosure of case information to opposing counsel based on the 1963 case of Brady v Maryland (373 US 83, 1963).

This experience helped me appreciate what terms like "stutter," "capillary electrophoresis," "allele dropout," and "pull-up" must seem like to lawyers and the general public. Likewise, when a scientist enters the legal arena, there are basic terminology and processes that must be understood in order to function in this environment. While not intended to be comprehensive in coverage, some important legal terms and concepts will be introduced in this chapter that should benefit DNA scientists going to court.

Terms and concepts that are second nature to forensic DNA scientists must be conveyed correctly in reports of analysis and in a court of law. Communication is a key factor. Thus, a discussion of jury comprehension is included in this chapter because in U.S. courts of law the expert testimony delivered by a DNA analyst needs to be understood by (typically) a dozen fellow citizens who have little-to-no science background. If this chapter and book can open the dialogue and help as a beginning translation between the different languages of science and the law, then I will have been successful.

Cultural Differences

As the majority decision in the landmark 1993 *Daubert* ruling notes (see opening quote for the chapter): "There are important differences between the quest for truth in the

TABLE 18.1 Differences in Culture between Science and the Law (Adapted from Bowen 2010, p. 33).

Issue	Science	Law
Truth	Serves the interest of itself	Serves the interest of the client
Communication	Open	Privileged
Process	Unbiased and systematic	Adversarial; defending a specific position
Goals	Provide socially valued goods and services; advance human knowledge; eliminate false beliefs; document knowledge through publication so that others can benefit	Serve the client; produce a better argument to the trier-of-fact than the opposing counsel

courtroom and the quest for truth in the laboratory." Table 18.1 contrasts some of these differences in culture between science and the law. Thus, when a scientist enters the culture of the courtroom to serve as an expert witness, he or she will probably feel out of place. As Clay Strange, a prosecutor in Travis County, Texas, has shared, "No amount of 'moot court' conducted by forensic scientists pretending to be judge, jury, prosecutor, and defense lawyer will fully prepare the expert for entry into the world of the criminal court. It has to be experienced to be believed" (Strange 2002).

As highlighted in Table 18.1, scientists are typically used to operating in an environment where information is shared in an open and hopefully unbiased fashion with a goal to advance human knowledge. Lawyers, particularly in a U.S. court environment, are adversarial meaning that they are arguing a particular position on behalf of a client.

For a criminal trial, the prosecution typically represents the government's position on behalf of a victim (or complainant) while the defense attorney represents a defendant who is being accused of having committed a crime against the alleged victim. Thus, neither side necessarily has the goal to find out "the truth;" however, all parties have an obligation to not permit a truly innocent person to be convicted.

Forensic science, on the other hand, is supposed to assist in finding out "the truth" through the collection and analysis of impartial data. However, because an individual's perspective, training, and experience can influence the interpretation of results, in some cases, conclusions may not always be impartial. The reality is that the ideal of finding "the truth" may not be reached by either science or the law. Nevertheless, hopefully everyone is doing their best in whatever role they have within the criminal justice system.

As Robin Bowen in her book *Ethics and the Practice of Forensic Science* has succinctly put it, "Attorneys are advocates, whereas experts [DNA scientists] are educators" (Bowen 2010, p. 36). Dr. Charlotte Word, an experienced expert witness, adds that. "It is the role of the expert witness to educate the trier-of-fact, who is either the jury in a jury trial or the judge in a hearing or a bench trial. It is often the role of the expert witness in pretrial meetings to educate the attorney asking questions on direct examination and to assist in the preparation of questions to be asked in the direct examination so that the qualifications, science, and evidence can be presented in a logical and coherent manner to the trier-of-fact. It is not the role

of the expert witness to take sides or be an advocate for the victim, the state or government, or the defendant. The expert witness is simply an advocate for good science" (CJW).

Right to Defense

Within the United States of America, defendants in criminal trials have a number of rights that are primarily declared by the Sixth and Fourteenth Amendments to the United States Constitution.

The Sixth Amendment, which was added with the original Bill of Rights in 1789, states:

> "In all criminal prosecutions, the accused shall enjoy the right to a speedy and public trial, by an impartial jury of the State and district wherein the crime shall have been committed, which district shall have been previously ascertained by law, and to be informed of the nature and cause of the accusation; to be confronted with the witnesses against him; to have compulsory process for obtaining witnesses in his favor, and to have the Assistance of Counsel for his defence" (see http://www.usconstitution.net/).

The Fourteenth Amendment, which was adopted in July 1868, states (in part):

> "All persons born or naturalized in the United States, and subject to the jurisdiction thereof, are citizens of the United States and of the State wherein they reside. No State shall make or enforce any law which shall abridge the privileges or immunities of citizens of the United States; nor shall any State deprive any person of life, liberty, or property, without **due process of law**; nor deny to any person within its jurisdiction the equal protection of the laws" (see http://www.usconstitution.net/; emphasis added).

In the context used here, the term "due process of law" refers to how and why laws are enforced for the protection of individual rights. As noted on the website USConsitution.net, due process generally guarantees (among other things): the right to a fair and public trial conducted in a competent manner, right to be present at the trial, right to an impartial jury, and right to be heard in one's own defense. A defendant is also permitted to know what evidence is being brought against him or her—a process known as *discovery*.

Discovery

Discovery is the legal process whereby information and materials relating to the prosecution of a defendant in an upcoming trial are shared between the prosecution team (e.g., law enforcement, crime laboratory, prosecuting attorney) and the defense counsel. The specific requirements of discovery are regulated by federal and state laws and, in some cases, the judge in a particular case may rule on the materials that must be provided during the discovery process. It is the obligation of the prosecution team to "turn over" to the defense attorney(s) all information and materials that relate to the prosecution of the defendant as well as any evidence or information that may be viewed as exculpatory (i.e., clearing of guilt) or that impeach (i.e., to dispute, disparage, or contradict) a prosecution witness. The defense generally has the responsibility to provide to the prosecution all reports and opinions from any experts expected to testify during the defense portion of the case. Additional information may also be provided by the defense, however the requirements tend to vary based on the jurisdiction and the type of proceeding.

Crime laboratory personnel are usually asked to provide the laboratory reports stating the opinions and conclusions from testing in a particular case as well as the basis for those opinions, which generally entails the complete laboratory case file with all supporting data. It is helpful if these materials are provided in a manner suitable for review by both sets of attorneys and other experts assisting either of the attorneys. Additional materials may also be requested from the crime laboratory depending on the jurisdiction and the challenges to the case. These materials may include laboratory standard operating procedures (SOPs), quality control manuals and/or records, training files, proficiency test records and files, audit documents including findings and corrective actions taken, validation studies, and publications from the laboratory.

LEGAL TERMS, CONCEPTS, AND IMPORTANT COURT CASES

Reviewing some legal terms, concepts, and court cases may help readers understand other aspects of this chapter.

Legal Terms and Concepts

***Voir dire*:** a French word for "to speak the truth" this term refers to the process of qualifying a juror or an expert witness to serve in court.

Trier-of-fact: the individual(s) who makes a ruling in court based on the evidence presented; for a jury trial, the jury is the trier-of-fact while for a bench trial or admissibility hearing the judge is the trier-of-fact.

Admissibility hearing: before specific evidence for a case can be heard by the trier-of-fact, the evidence must be deemed sufficiently reliable to be of value for presentation in court; the judge may decide the evidence is generally accepted (because it has been used in previous court cases) or may require a separate hearing with arguments for and against the evidence being admitted into court.

Custodian of records: Under the Federal Laws of Evidence (see below), a "custodian of records," who is, for example, a DNA analyst from a laboratory with knowledge of the lab's records and the specific case report and its preparation, is permitted to testify to records prepared in the routine course of business. An individual who meets the requirements of custodian of records may be permitted to testify to the records and findings in a case once the appropriate legal foundation for the testimony under these rules are established and meet the requirements of the court. A DNA laboratory report and case file are typically considered the "business records." However, after the recent case of *Melendez-Diaz v. Massachusetts* (see below), lab reports may no longer qualify as "business records."

Care in discussing original source of reference sample in a cold database hit: Connecting an individual to a crime through a DNA database cold hit can complicate things in the courtroom. Attorneys have to be very careful not to mention that the defendant's DNA profile was on a "convicted offender" index within a DNA database as this could be prejudicial to the defendant in the jurors' minds to know that the defendant has a prior criminal record. A mistrial could result from mentioning that a prior criminal record is the reason the individual's DNA profile was in the database that led to connecting him or her to

the crime under consideration. Sometimes testimony may be given though to matching on "a database" with no mention of the "convicted offender" portion of the database.

Important U.S. Court Cases

Below are a few of the more prominent cases that impact scientific evidence and expert witnesses. Several of these cases will be covered in more detail later in the chapter.

***Frye v. United States*:** a 1923 Court of Appeals case establishing the initial legal precedence for admissibility of scientific evidence

***Brady v. State of Maryland*:** a 1963 Supreme Court case requiring that the prosecution must disclose any exculpatory evidence to the defense

***Gideon v. Wainwright*:** a 1963 Supreme Court case that established the right to defense counsel (public defender) in state court proceedings for indigents accused of serious crimes

***Daubert v. Merrell-Dow Pharmaceuticals*:** a 1993 Supreme Court case making the trial judge a gatekeeper to scientific evidence admissibility

***Crawford v. Washington*:** a 2004 Supreme Court case strengthening defendants' rights to confront and cross-examine witnesses against them

***Melendez-Diaz v. Massachusetts*:** a 2009 Supreme Court case requiring that submission of a forensic laboratory report in court requires the testimony of the person who performed the test

THE COURTROOM SCENE

While different state-specific laws mean that some aspects of trials vary between states and legal jurisdictions, the courtroom scene is generally the same throughout the United States and other countries formerly under British rule. An effort was made to keep the material presented here as general as possible while providing specific details where deemed appropriate.

The actions of a court of law are somewhat analogous to a scripted play conducted in a theater. Just as in a major theatrical production, in courts there are actors with scripts to be followed who take specific actions at specific times, extensive preparation before the actual performance, events that play out behind the "curtains," drama for the defendant who has a major stake in the outcome, and oversight by a director. The following sections will discuss the personnel or "actors" involved in a trial along with their individual responsibilities, the actions taken as the courtroom proceedings unfold, and the preparation that precedes the actual trial.

Although this analogy to a theatrical production may help to provide better understanding towards the actions of a court of law, it is important to keep in mind that court proceedings are very serious and can have a major impact on the lives of defendants, victims, and their families.

Courtroom Proceedings

For a jury trial involving a criminal case in the United States, a jury, composed of a sufficient number of members of the general public (typically 12) and often including alternate members of the jury, is first selected. While a judge presides over the trial, the selected jury

will be responsible for coming to a conclusion regarding the individual(s) on trial—either "guilty" or "not guilty"—based on the evidence presented.

Opening statements are made by both the prosecution and the defense outlining the issues in the trial. After opening statements have been made, the prosecution, representing the government's position on behalf of the victim(s), presents its case by putting witnesses before the jury and entering items of evidence. The prosecution has the burden of proof to establish beyond a reasonable doubt the guilt of the defendant(s).

After the prosecution has made its case, the defense team may, but is not required to, introduce additional witnesses to try to cast doubt on the prosecution's case. Closing arguments are offered by both sides at the conclusion of the trial. The jury then considers the evidence presented and testimony heard and renders a verdict of "guilty" or "not guilty." In a criminal trial, for a defendant to be convicted, the jury must unanimously reach a "guilty" verdict based on "proof beyond a reasonable doubt." In a civil trial, although the jury's verdict must typically also be unanimous in favor of one party, the "burden of proof" is somewhat lessened requiring only a "preponderance of evidence."

For each expert witness who is part of a trial, there are five parts to the testimony after being called to the witness stand and sworn in: (1) establishing the witness's qualifications or *voir dire*, (2) direct examination, (3) cross-examination, (4) re-direct and re-cross-examination, and (5) the judge retaining or excusing the witness. Expert witnesses may also be called to testify during a preliminary hearing.

Courtroom Personnel

Like a set in a play or movie, there are a number of actors who each have a specific role in the production of a trial. These include the judge, the court reporter, the plaintiff(s), the defendant(s), the prosecuting attorney(s), the defense attorney(s), the jury, and the expert witnesses or other witnesses called to testify in the witness box/stand.

The **judge** oversees the proceedings of the trial. He or she makes decisions about admissibility of evidence and decides on objections raised by lawyers during the course of the trial. The judge's aim is to ensure that both sides receive a fair trial and that only legally admissible evidence is presented to the trier-of-fact. The judge is always mindful that a conviction may be followed by an appeal, and that the appellate court will scrutinize the evidence to determine if the defendant received a fair trial. For that reason, the judge will pay close attention to the "record," including whether the court reporter can produce an accurate transcript of the expert witness testimony.

The **court reporter** creates a transcript of the court proceedings. Every word is recorded so that this information may be reviewed if necessary. What the court reporter writes becomes the facts of the case so it is important for witnesses to speak clearly.

The **prosecutor** seeks justice on behalf of the plantiff or complainant. Ted Hunt, a prosecutor from Kansas City, Missouri, remarks: "The prosecutor's role in the criminal justice system is to seek justice in each case rather than to merely convict. In many instances, seeking justice equates with the aggressive prosecution of a criminal defendant. In other cases, however, seeking justice might mean dismissing a case because the evidence of guilt is in doubt. It may also mean reducing a charge when the evidence reveals that a defendant's degree of culpability is less than previously believed" (TRH).

The Orange County (California) District Attorney's DNA Unit adds: "A prosecuting attorney has an obligation and duty to see that 'justice is done." This obligation extends well beyond merely convicting the guilty. It includes ensuring that no one is unfairly convicted and that the innocent are exonerated:

> [The prosecutor] is the representative not of an ordinary party to a controversy, but of a sovereignty whose obligation to govern impartially is as compelling as its obligation to govern at all; and whose interest, therefore, in a criminal prosecution is not that it shall win a case, but that justice shall be done. As such, he is in a peculiar and very definite sense the servant of the law, the twofold aim of which is that guilt shall not escape or innocence suffer. He may prosecute with earnestness and vigor — indeed, he should do so. But, while he may strike hard blows, he is not at liberty to strike foul ones. It is as much his duty to refrain from improper methods calculated to produce a wrongful conviction as it is to use every legitimate means to bring about a just one (*Berger v. United States* (1935) 295 U.S. 78, 88)." (OCDA).

The **defense attorney** plays a key role in the criminal justice system on behalf of a defendant. Brad Bannon, a defense attorney in Raleigh, North Carolina, shares: "The role of a defense attorney is to provide independent and zealous representation to people accused of crimes, and to give as much meaning as possible to the presumption of innocence and the burden of proof that the government must meet before it may limit or deprive a person of freedom. First, you review the prosecution's evidence and independently pursue as much evidence as you can and should in each case. After evaluating all of that evidence in light of your practical and legal knowledge and experience, you provide your best advice to the accused about what he or she should do, whether that involves going to trial or exploring resolutions outside of trial, such as a plea agreement. Finally, if the case ends up in a contested forum, you do your absolute best to make sure the prosecution is put to its constitutional burdens in that forum" (BB).

An **expert witness** (e.g., a DNA analyst) is called to testify in court regarding data collected and conclusions drawn based on that data. Law professor K. A. Philipps once wrote: "Just as the lawyer has a job to do in the courtroom, so does the scientist. The job of the forensic expert is only half done after the scientific tests have been done and the written reports have been prepared. The other half is to present the results of the tests in a form that is legally admissible and in a manner that is understandable to the judge and jury" (Philipps 1977).

Most experts presenting testimony regarding DNA test results and conclusions will be called to court by the prosecution to provide some link of the evidence from a crime to a victim or to the defendant. Most crime laboratory personnel never have the opportunity to be called by the defense since generally inconclusive results and/or results that excluded other suspects are not presented in court and are generally of no or limited value to the defense. In some cases, experts may be called by the defense to provide differing opinions or challenges to the DNA testing. The processes described below are the same regardless of whether the expert witness is being called to testify in a case by the prosecution or the defense.

Preparation of the Expert Witness

As with any performance, prior preparation is vital for success. Additional thoughts on details of this preparation will be covered later in the chapter.

Ted Hunt offers a prosecutor's perspective: "Perhaps the most vital aspect of testimonial preparation is [the witness] meeting with the prosecutor for a pretrial conference. If at all possible, this conference should be done in person, rather than over the phone. At this meeting, the prosecutor should ask the expert what issues or anomalies are present in the evidence, case data, or test results. The prosecutor should also review his or her anticipated questions and areas of inquiry with the expert and receive his or her feedback. It's also helpful for the expert to suggest issues or topics to highlight, de-emphasize, or exclude. In addition, the prosecutor and the expert witness should collaborate to determine what, if any, visual aids may help enhance the jury's comprehension and retention of the evidence. Finally, an informal 'dry-run' of the expert's direct examination or anticipated cross-examination during the pretrial conference may also help the prosecutor and the expert enhance their courtroom presentations" (TRH).

Charlotte Word shares her opinion as a frequent expert witness: "It is important for the expert witness and the attorney to meet prior to testifying. This provides an excellent opportunity to discuss if all discovery has been supplied, ensure that the attorney understands your limitations and the limitations of the testing, and to draft and/or discuss the series of questions that will be asked in court. Having a pretrial meeting helps ensure that the attorney covers all of the relevant qualifications of the witness, scientific parameters of the test and of the specific test results and conclusions. Neither the attorney nor witness should be 'blind-sided' with surprise questions or answers at the trial. Remember that it is the responsibility of the expert witness to educate the jury by providing them with sufficient information to understand the meaning of the test results obtained in the laboratory as well as have a good sense of the quality of the work performed and the credibility of the witness and the crime laboratory. This can best be achieved by pretrial preparation. If the attorney has never been to the laboratory, it may be helpful to invite the attorney to the laboratory for a tour and more detailed explanations" (CJW).

Courtroom Performance of the Expert Witness

Swearing in and Offering Testimony

Raymond Davis in his Courtroom Skills training course on expert testimony offers: "Pay attention to the little details. For example, if you carry your file or briefcase in your left hand, then you will be able to easily raise your right hand to be sworn in as you enter the witness stand." Remember that anything you take to the stand can be examined by both sides, including any personal notes used to assist with testimony.

A transcript is prepared by court reporters who attempt to capture every word stated by the lawyers and the witnesses as part of the court proceedings. In a sense, the "script" for the courtroom "stage performance" is produced after the performance is over. This transcript provides a written record of every word and can be reviewed to confirm previous aspects of the trial during the trial or used following the trial as a complete record of the events that transpired.

Speak slowly and simply. If there is a difficult word, define the word in lay terms for the jury and spell it out to make life easier for the court reporter. The judge is the "director of the performance" and is in charge. The witness is a member of the "performance cast" and may not leave the witness stand or the courtroom until excused by the judge.

*Qualification (*Voir dire*)*

Before an expert is allowed to offer testimony in court, he or she must be qualified by the court to speak on the topic at hand in what is known as *voir dire*, which as stated earlier is French for "to speak the truth." To demonstrate the qualifications of the expert witness, the attorney who called the witness to testify asks questions detailing the expert's education, training, and experience. Some example questions are included in Table 18.2. This testimony lays the foundation for allowing the expert to testify to the analysis performed in the laboratory and to the results and conclusions derived.

A copy of the expert's *curriculum vitae* or CV is generally provided to opposing counsel, and may also be entered into the court's record during testimony. The CV contains educational background, experience in current and previous positions, training, and a listing of any professional societies of which the witness is a member along with any scientific articles published. According to Dr. Robin Cotton, who has testified in more than 200 trials: "Your CV should be only the organizations you are a member of, your publications, your job title, your job description, and your education. Testimony is not a qualification, neither is your volunteer work or anything else" (RWC). Some attorneys will use the expert's CV to help craft questions covered as part of the *voir dire* process.

The opposing counsel has the opportunity to ask additional questions regarding the witness's qualifications either during the *voir dire* or during cross-examination. In some jurisdictions, the judge makes a formal acceptance of the witness as an expert in a particular scientific field (e.g., expert in forensic DNA testing).

Direct Examination

As noted in the court proceedings section above, following *voir dire* and the witness being qualified to provide expert testimony, direct examination will begin. Some common questions asked during a direct examination are listed in Table 18.3.

There is not likely to be an ideal direct examination because details to a particular case will be case specific. However, as noted by Robin Cotton: "Lawyers like it when you have a list of prepared questions for your direct examination. However, you need to go through the

TABLE 18.2 Some Potential Questions during Qualifying the Expert (*Voir Dire*) Courtesy of Bruce Heidebrecht (DNA Technical Leader, Maryland State Police Crime Laboratory).

1) Please state your name and spell it for the court.
2) Where are you currently employed?
3) How long have you been employed at the Crime Laboratory?
4) What is your current job position?
5) What are your current job duties?
6) Prior to working at the Crime Laboratory, what was your previous work experience?
7) What is your educational background?
8) What type of training have you had in the field of forensic DNA analysis?
9) Have you previously been qualified as an expert in the field of forensic DNA analysis? How many times and in which courts?

TABLE 18.3 Some Potential Questions during a Direct Examination Courtesy of Bruce Heidebrecht (DNA Technical Leader, Maryland State Police Crime Laboratory).

1) What is DNA?
2) Where is DNA found in humans?
3) Will two different people have the same DNA?
4) What type of DNA testing was performed in this case?
5) Can you briefly explain what steps are involved in STR analysis?
6) Is STR analysis a new technology?
7) Why has the forensic community adopted the use of STR analysis?
8) Is the field of forensic science the only area where STR analysis is utilized?
9) Are all of the procedures used at the Crime Laboratory generally accepted in the scientific community?
10) Have STR analysis results been accepted in the courts in this state, and in other states?
11) What safeguards and controls are in place at the Crime Laboratory to ensure the integrity of the testing procedures and the integrity of the DNA results?
12) If any of these controls do not work properly, what conclusions will be made?
13) Did there come a time when the Crime Laboratory was requested to conduct STR analysis in the investigation of _____?
14) What evidence was submitted to the DNA unit for analysis?
15) Which of these samples were then subjected to STR analysis?
16) Were results obtained from all of the evidence tested?
17) Were you able to reach a conclusion based on the STR analysis of this evidence? And if yes, what are the conclusions?
18) Were you able to reach a conclusion as to the frequency of this DNA profile occurring in the general population? And if yes, what are those conclusions?
19) After analysis is completed, are your case files subject to peer review?

questions with them because some of the questions may not fit a particular case or your role in the expert testimony you are offering" (RWC).

Ted Hunt shares his perspective as a seasoned prosecutor: "I have an outline of the facts I need to establish or the topics I need to cover, but I don't write out my questions. I believe that writing down questions verbatim locks you into a script that takes away the flexibility needed during witness examinations. Glancing down at a topic or an anticipated answer rather than a question gives me the flexibility needed to formulate a question that will elicit the desired information. If you get out of sync with an expert in court based on a prepared script, both prosecutor and expert are going to get lost and will look unprepared to the jury"(TRH).

Explaining DNA Results to a Jury

The concepts of DNA can be challenging for lay jurors to comprehend. Brad Bannon advises expert witnesses: "Understand that you are not lecturing to your professional colleagues. You are speaking to people who will likely have little or no experience in your area of expertise, but you must effectively share that expertise with them without their eyes glazing over. Speak in language that is accessible but not condescending" (BB).

Charlotte Word recommends: "Just show a final table of results and conclusions rather than raw data. However, some attorneys may want to present the actual electropherograms. This

may be necessary to lay appropriate background to expected opposing counsel challenges (e.g., is the allele major or minor, is the peak a stutter product or an allele, etc.)" (CJW).

When Ted Hunt was asked how much detail should be shared with the jury, he replied: "I think Albert Einstein summed it up best when he said, "Everything should be made as simple as possible, but not simpler.' The jury needs just enough information to accomplish its assigned task—to determine whether or not the DNA test results in question are relevant, reliable, and persuasive in the context of a given case" (TRH).

Cross Examination

Cross examination, where the opposing counsel is trying to discredit the testimony of the witness, can be particularly grueling and challenging for the witness. When asked what she enjoyed least about being an expert witness, Robin Cotton said, "Very aggressive, very loud, very unpleasant cross-examination tactics. I don't personally care to be shouted at." She astutely observes: "When somebody is attacking your testimony, there are two bad things that happen. One is that you get mad. The other thing is you get tired so you tend to start giving in or you just shut down or you glare at everybody and you look uncomfortable. It can take some practice at home, such as rehearsing how you might respond to challenging questions" (RWC).

In spite of how a witness is treated by the opposing counsel, Dr. Cotton continues: "You should not treat the prosecution and defense attorneys any different in how you respond to them. However you talk to one attorney, you need to talk to the other attorney the exact same way. You don't want to change your body language. You don't want to change your tone or the speed in which you answer questions or your demeanor to the jury" (RWC).

She goes on: "It is very hard not to take a tough cross-examination personally. Anytime you feel like you are being attacked then your anxiety goes up, your adrenalin goes up, and you feel like you have to defend yourself. I think that is just part of the tactic. You are not being attacked really. You are there just to say, "This is the data." And the defense attorneys are there to say, "Are you sure, did you do it right?" That is their job. If you didn't do your work right, then they need to know! It is their job to discuss anything that may be incorrect in what you did. They want and need to point it out" (RWC).

Being cross-examined can be exhausting and emotionally taxing. Dr. Cotton shares, "I learned how to fix my feelings of frustration after participating in the courtroom drama by remembering that it was a show. *As an expert witness, I have a very important role in this show—and the show has very important consequences—but the show isn't my permanent world*. I have strengths and other things outside of this arena. Whatever happens in the courtroom, I don't have to carry it out with me."

We have briefly covered courtroom proceedings and personnel and explored the preparation and performance expected of an expert witness in a court of law. The next section examines the role of a scientific expert in more detail.

THE SCIENTIFIC EXPERT WITNESS

In 1901, a future federal judge named Learned Hand from Albany, NY, published an oft-quoted treatise in the *Harvard Law Review* on expert witnesses. He begins his article: "No

one will deny that the law should in some way effectively use expert knowledge wherever it will aid in settling disputes. The only question is as to how it can do so best…" Judge Hand expressed concern over use of expert witnesses because (1) they are typically "a hired champion of one side" in a case and thus can lose objectivity and (2) dueling experts are permitted in the court proceedings that may contradict one another and thus confuse the jury (Hand 1901, p.53). He feared that the exchange of technical arguments would not make sense to the jury and the more charismatic of the experts could influence a jury's decision rather than appropriate scientific issues.

An expert is someone who has specialized knowledge on a particular topic based on education, training, experience, or skill. Based on this education, training, experience, and/or skill, the expert knows more than a lay person on a particular subject. Thus, the expert is called upon to use this specialized knowledge to enlighten the court of law.

Role of the Expert Witness

Robin Bowen notes that "expert witnesses are held in high esteem by the jury, so it is important to provide information accurately and authoritatively" (Bowen 2010, p. 64). In addressing the role of an expert witness, Robin Cotton states: "Your job is to represent your data. I think people forget that usually just one person signs the report. I think that it is important that many people in the lab would have the same opinion about that report. When you are sitting there on the witness stand, you are the lab! For purposes of that jury, you are the laboratory. Whatever they think about the laboratory comes from what they think about you… Doing a poor job with courtroom testimony, can cause serious problems for a laboratory and the analyst. If laboratory personnel are testifying all of the time, then every one of those people has to be good every time" (RWC).

Responsibilities of the Expert Witness

Robin Cotton notes: "You have to be really focused in the courtroom. You cannot mentally be somewhere else. You need to listen to those questions. The longer you are listening to questions, the harder it is to stay focused. Just like anything, you get tired. The attorneys get to take a little rest, they trade off, etc., but you cannot get tired. You have to be very focused. You cannot be paying attention to everything in the courtroom. It will distract you. I just look to the attorney and I just look to the jury and that's it" (RWC).

Some suggestions and informal rules (distilled from Bowen 2010, p. 63):

- Do not discuss the case with anyone outside of court including other witnesses. "In some jurisdictions, the expert is not permitted to discuss the case or testimony with either attorney once the testimony has begun" (CJW).
- Avoid advocating one position or another.
- Do not be defensive or argumentative in responses to questions.
- Stop talking if interrupted by the judge or an attorney (respect courtroom protocol).
- Request permission to answer in greater detail if something beyond a "yes/no" answer is needed to clarify a response.
- Do not avoid a question if you believe your answer will support the opposing side (credibility will be negatively impacted if a bias is shown or there is hesitation in providing an objective answer).

- Do not volunteer information without being asked.
- Do not guess.
- Do not generalize answers.
- It is acceptable to answer with "I do not know – but if I looked at my case notes, it would refresh my memory" (RWC).
- Make sure that you fully understand the question before answering.
- Present opinions truthfully after all information is collected and examinations are completed.
- "If you make or find a mistake, correct it. Do not let errors remain uncorrected for the jury or the court record" (CJW).

Think before you speak. You may be asked challenging hypothetical questions regarding the conclusions made. Pay close attention to the question asked and only answer the question.

Robin Cotton advocates that as an expert witness you should have "unconditional positive regard" for everyone in the courtroom. She emphasizes: "in that arena of the courtroom, they all get my equal respect genuinely. I think that is where the creditability comes from that you should have as an expert witness" (RWC).

Trial Preparation with Attorneys

Prior preparation prevents poor performance (the 5 P's). The expert witness and attorney must work closely together to obtain optimal performance in the courtroom. The expert should have carefully reviewed the entire case file and report especially if the work was performed several years before. Refresh your memory of the details. Work through the calculations to make sure you understand all of them. Re-review the results and conclusions to be sure everything was reported correctly.

In a pretrial conference find out if the attorney assigned to the case has handled DNA evidence previously. If not, then invite the attorney for a lab visit to help familiarize him/her with the steps involved in DNA analysis. This can improve communication. Offer a prepared list of questions that can be used for *voir dire* (see Table 18.2) and direct examination (see Table 18.3).

When an expert witness cannot appear in person, an affidavit can be provided. An *affidavit* is a written declaration or statement of facts made voluntarily and confirmed by oath. Ted Hunt comments: "After the *Melendez-Diaz* decision, affidavits can no longer be provided in lieu of live testimony at trial. However, most courts have held that the sixth Amendment right to confrontation does not apply at pretrial or preliminary hearings. The U.S. Supreme Court has not yet definitively spoken on this issue" (TRH).

Some jurisdictions utilize a *deposition* where opposing counsel formally asks an expert witness questions prior to trial. This approach is generally only used for civil proceedings. While the proceedings are done without a judge being present, it is part of the court record and a transcript is made of all questions and responses. Charlotte Word emphasizes: "Preparation for the deposition should be as thorough as for the trial. If the deposition transcript is available, the witness should read it prior to testifying in the trial taking special note of any discrepancies between what was said versus what was recorded and noting any information that might have changed since the time of the deposition. The attorney and the

expert witness should be prepared to address any differences between what is present in the transcript and what will be said at the trial as this should be an obvious expected challenge from opposing counsel" (CJW).

Communication in Court

An expert witness is a teacher, an educator. The purpose of being present in a court of law is to explain scientific results under direct examination and to defend them when cross-examined by the opposing counsel. In a jury trial, the expert witness should face the jury and address them as they are the decision makers (the "triers-of-fact") needing the expert's input.

Simple visuals in a "show and tell" format can be effective in many circumstances. However, Charlotte Word reminds experts: "Be sure any presentations are coordinated between the attorney and the witness. Be sure to preview and carefully check for accuracy any exhibits related to your testimony that the attorney has prepared" (CJW).

Most modern courtrooms have PowerPoint projection capability. Keep in mind though that everything you present should have a specific purpose. You are not in court to give a genetics lecture—nor would the judge and opposing counsel likely permit you to pontificate endlessly without addressing a specific question!

Ted Hunt states: "The best advice I have for any speaker, expert DNA witnesses included, is to know your audience and speak to them in their native tongue. A brilliant analyst can make a lousy expert witness when he or she fails to connect with the jury. *Many times, this failure to connect stems from the fact that the expert testifies as if he or she is delivering a scientific dissertation to a room full of colleagues rather than teaching a classroom full of novices*. Given the time and procedural constraints of the courtroom, the use of scientific jargon, industry acronyms, and excessively complicated explanations creates cognitive barriers for jurors that are difficult to overcome" (TRH).

Several years ago, two Los Angeles County prosecutors asked the question, "How do you prepare to testify in front of twelve average citizens?" They answered their own question with: "One way to assess your ability is to practice on friends who have no scientific background. Explain DNA testing to them simply and see how they respond. This will give you a sense of whether you are reaching your audience. It is also helpful to be familiar with the basic content of most expert testimony and to review these areas in advance each time you testify" (Kahn & Feldstern 1998). Robin Cotton agrees: "Go explain [DNA] to your mother until she understands or father or brother or whatever. It does take a lot of practice to say it without using your normal language, your normal scientific terms" (RWC).

Credibility

Your actions both on and off the stand impact the perceptions that others will have of you. Be professional. Act confident but not condescending. Your credibility as an expert witness can be impacted by your attitude and demeanor. Robin Cotton explains: "The minute you walk into the courthouse you need to be aware that anyone around you could be a juror, an attorney, or someone else involved in the trial. Your professional behavior starts when you walk in the door and you don't change that until you are far away from that courtroom" (RWC).

Gil Sapir offers 12 points to help maintain credibility during court testimony: (1) be nervous, (2) always tell the truth, (3) listen to the question, (4) pause, then answer, (5) admit mistakes and problems, (6) admit limitations, (7) admit inability to remember, (8) do not hedge or obfuscate, (9) speak to the jury, (10) maintain a consistent attitude, (11) never argue with counsel, and (12) answer just the question (Sapir 2002). Concerning the above recommendation to be nervous, Charlotte Word shares: "This is an excellent set of recommendations, although I question the need to be nervous. Being well prepared should alleviate most nerves. However, as before any presentation, it is always helpful to be alert and perhaps have a small jolt of adrenaline" (CJW).

Having read a number of trial court transcripts involving DNA testimony, not all expert witnesses follow these recommendations in many cases so there is room for improvement by even experienced experts! One last point under credibility: keep your CV up-to-date if you testify regularly as you should continually have attendance at meetings or training workshops to include.

Dress and Appearance

How you dress will impact what others think of you and either add or detract from your credibility. When you look your best you will also act more professionally. A suit and tie are appropriate for men and a business suit or dress and jacket fitting for women. Speaking on how she dresses for court, Robin Cotton states: "I wear dark clothes. I wear jewelry but I don't wear anything big. I don't wear a red suit. I don't look like I am going to a party. Nobody in that courtroom is there because something good happened, and so you have to respect that things have happened and you have to have respect for the proceeding and you have to have respect for the individuals who are there" (RWC).

In terms of what she takes into the courtroom, Dr. Cotton relates: "I take whatever I think I am going to need to the witness box. If there were a *Frye* hearing, I might bring in scientific articles. For a case, I might bring some notes along with my copy of the case folder. I don't have things with me that I'm not willing to give over. Some people take a book if they think they might be waiting—and that is fine. I usually take some scientific journal articles just because *People* magazine doesn't seem to be the right thing to have in a courtroom" (RWC).

Charlotte Word adds: "The case file may be carried into court for reference, but if you are taking additional materials (e.g., references, SOPs, etc.) a small business folder or briefcase is appropriate to use. Ladies may carry small professional handbags, as needed, but if possible, all coats, umbrellas, large handbags, luggage, and any other accessories not needed to enhance the testimony should be left in the car, the attorney's office, or any other appropriate place. Care should be taken to stay professional and in professional attire anywhere near the court where the jury has a chance of seeing you; changing into work or travel clothes should take place away from the courthouse" (CJW).

Ethics

Ethics is commonly defined as a set of moral values or precepts that govern rules of conduct followed by an individual. Expert witnesses need to be ethical in their scientific work and in their court presentation. An expert witness must be impartial and not have a

professional interest in the outcome of the trial. You should be true to the evidence and the science NOT to either the prosecution's or the defense's position. The advice, attributed to President Harry S Truman, to "Always tell the truth; that way you never have to remember what you said" applies to forensic scientists in their role as expert witnesses.

Charlotte Word comments: "Providing unbiased and neutral testimony is the best and easiest position for an expert witness to take. If an expert can answer the question, 'Would my answer be the same if it were asked by the other attorney in this case?' with a resounding 'Yes!,' then the testimony is likely to be impartial. The expert has no obligation to answer a question the way one of the attorneys requests particularly if the answer is disingenuous in any manner" (CJW).

Robin Bowen in her book *Ethics and the Practice of Forensic Science* provides several challenges expert witnesses may face including:

- Resisting attorneys who want testimony that supports their client's position
- Producing evaluations that prove disadvantageous to the side that has retained them
- Being asked by an attorney to not write a report which would then be discoverable and possibly help the opposing attorney
- Having opinions distorted and their reputation impugned (Bowen 2010, pp. 38–89).

The various professional associations of forensic scientists have codes of ethics and conduct. For example, the American Academy of Forensic Sciences Bylaws Article II, section 1(c) states: "No member or affiliate of the Academy shall materially misrepresent data or scientific principles upon which his or her conclusion or professional opinion is based" (http://www.aafs.org/aafs-bylaws). The American Society of Crime Laboratory Directors Laboratory Accreditation Board (ASCLD/LAB) has developed guiding principles for forensic scientists and their management. They are available at http://www.ascld-lab.org/about_us/guidingprinciples.html and cover ethics and professionalism, competency and proficiency, and the importance of clear communication in court testimony.

ADMISSIBILITY OF EVIDENCE

Standard DNA tests are widely used and deemed admissible by courts around the world. However, as new tests are developed and implemented, their results must be deemed admissible in court. Charlotte Word comments: "I think it is still important to understand what an admissibility hearing is all about so that one can be prepared if challenges to the testing procedures are raised" (CJW). In this section, several important court cases regarding admissibility of scientific evidence and testimony are reviewed.

The *Frye* General Acceptance Standard

A landmark court case in 1923 established the initial legal precedence for admissibility of scientific evidence in a U.S. court of law. *Frye v. United States* was a Court of Appeals of the District of Columbia decision rendered December 3, 1923. It was based on an appeal from a lower court's refusal to admit test results from a systolic blood pressure device (a polygraph or "lie detector"). James Alphonzo Frye, convicted for second degree murder and trying to

use his passing of a lie detector test to show he did not commit the crime, has achieved a level of legal immortality through lending his last name to criteria for accepting scientific evidence in court. In the end, the lie detector results were deemed inadmissible and Mr. Frye's conviction was upheld.

In its now famous ruling, the court crafted a "general acceptance" standard to determine whether scientific evidence was sufficiently reliable to warrant consideration by a trier-of-fact:

> "Just when a scientific principle or discovery crosses the line between the experimental and demonstrable stages is difficult to define. Somewhere in this twilight zone the evidential forces of the principle must be recognized, and while courts will go a long way in admitting expert testimony deduced from a well-recognized scientific principle of discovery, the thing from which the deduction is made must be sufficiently established to have gained general acceptance in the particular field in which it belongs" (*Frye v. United States*, 54 App. D.C. 46, 293 F. 1013 [1923]; emphasis added).

The general acceptance criterion has often been interpreted to mean the science of the underlying principle has been published in a peer-reviewed scientific journal and is used and accepted by other scientists in general, as well as scientists in the "relevant" field. It must also show that the general technology has been correctly applied to forensic evidence. In jurisdictions following the Frye general acceptance standard, a "Frye hearing" may be held prior to the trial in order for the judge to assess whether or not the scientific evidence should be admitted into court. These hearings can sometimes be quite contentious and involve experts brought in to represent both sides to the argument.

Although most jurisdictions in the United States have moved to a more recent standard for assessing scientific evidence (the following section on the *Daubert* ruling), states still following the *Frye* standard include California, Florida, Illinois, Kansas, Maryland, Michigan, Minnesota, Missouri, New Jersey, New York, Pennsylvania, and Washington (http://en.wikipedia.org/wiki/Frye_standard).

Federal Rules of Evidence

In the United States, whether or not evidence may be admitted to a court of law is controlled by federal or state rules of evidence (depending on the court). The Federal Rules of Evidence (FRE) were originally enacted in 1975 by Public Law 93–595 and have been amended periodically by Congress or the U.S. Supreme Court over the years (for an up-to-date version, see http://www.federalevidence.com).

There are 11 articles in the FRE making it similar in format to the U.S. Constitution. Article VII deals with opinions and expert testimony and contains six rules (Rule 701 through Rule 706). FRE Rule 702 governs "Testimony by Experts" and is the primary rule pertinent to expert witnesses:

> If scientific, technical, or other specialized knowledge will assist the trier-of-fact to understand the evidence or to determine a fact in issue, a witness qualified as an expert by knowledge, skill, experience, training, or education, may testify thereto in the form of an opinion or otherwise, if (1) the testimony is based upon sufficient facts or data, (2) the testimony is the product of reliable principles and methods, and (3) the witness has applied the principles and methods reliably to the facts of the case. (As amended Apr. 17, 2000, eff. Dec. 1, 2000.)

FRE Rule 702 opens the door for judges to permit expert testimony if it assists the trier-of-fact (e.g., the jury in a jury trial) in understanding evidence. Several other Federal Rules of Evidence also pertain to witnesses and evidence including Rule 104(a) which gives the trial judge the power to determine the admissibility of the evidence and expert witnesses; Rule 401, which defines "relevant evidence"; Rule 402, which allows all relevant evidence to be admissible; Rule 403, which permits relevant evidence to be excluded from the trial if it is deemed to be a waste of time or to be misleading to the jury; and Rule 615, which gives the trial judge the authority to exclude a witness so that he or she cannot hear the testimony of other witnesses in the trial.

States have similar rules of evidence that typically adopt the FRE in regards to the "testimony by experts" section. Links to state rules of evidence can be found on the National Clearing House for Science, Technology, and the Law website: http://www.ncstl.org/resources/702.

California's Kelly/Frye General Acceptance

A California Supreme Court case in 1976 of *People v. Kelly* (549 P.2d 1240 [Cal 1976]) has shaped California's legal landscape for admissibility and builds upon the *Frye* general acceptance. Kelly/Frye requires a preliminary showing of general acceptance of the new technique in the relevant scientific community followed by establishment of method reliability based on testimony from a qualified expert. California still applies its *Kelly/Frye* standard for admissibility of expert testimony despite the 1993 U.S. Supreme Court *Daubert* decision.

The *Daubert* Ruling

In 1993, 70 years after the *Frye* standard was set, the U.S. Supreme Court ruled that the Federal Rules of Evidence, which had been enacted in 1975, should supersede *Frye* in their *Daubert v. Merrell-Dow Pharmaceuticals* case (509 U.S. 579 [1993]). This case involved the parents of two children, Jason Daubert and Eric Schuller, born with birth defects that they claimed came from taking a morning sickness drug named Bendectin produced by Merrell-Dow Pharmaceuticals. The plaintiffs in this civil trial offered eight different experts with animal studies, chemical structure analysis, and other studies generated for specific use in this court case that had not previously been peer-reviewed and published in the scientific literature. The original trial court had barred this expert testimony and Daubert's lawyers appealed the case to the U.S. Supreme Court.

In a 7-2 decision, the Supreme Court ruled against the plantiffs (agreeing with the original court ruling) and offered a new standard for admissibility of expert witness testimony during federal legal proceedings where the trial judge is permitted to act as a "gatekeeper" to scientific evidence admissibility. *Daubert* thus provided greater flexibility over the "general acceptance" criteria of *Frye*.

The U.S. Supreme Court sent the *Daubert* case back to the Ninth Circuit Court of Appeal, which two years later sustained the original trial court ruling barring the plantiff's experts' opinions (*Daubert v. Merrell Dow Pharmaceuticals* 43 F. 3d 1311 9th Cir. 1995]). This court ruling concluded: "…in determining whether proposed expert testimony amounts to good science, we may not ignore the fact that a scientist's normal workplace is the lab or the field, not the courtroom or the lawyer's office" (p. 1317).

There are two "prongs" or portions to the *Daubert* decision—namely is the scientific evidence and testimony offered by the expert witness both (1) *reliable* and (2) *relevant* to the case at hand. Five factors are described in this decision in helping the judge determine whether there is a scientifically valid and reliable foundation for the evidence. These include: (1) the theory or technique must be falsifiable, refutable, and testable, or in other words involve the scientific method, (2) the theory or technique has been subjected to peer review and publication, or in other words submitted "to the scrutiny of the scientific community," (3) the court should consider the known or potential rate of error with the theory or technique, (4) standards or controls for the theory or technique exist and are maintained, and (5) the theory or technique is generally accepted in the relevant scientific community. Thus, the *Frye* standard is actually included within the *Daubert* ruling albeit with further details.

With the second *Daubert* prong of "relevance," the trial court must decide if the methodology can be properly applied to the facts under consideration. In other words, is the method appropriate or "fit for purpose" in the case at hand? The Supreme Court decision in *Daubert* emphasizes scientific validity with a focus "on principles and methodology, not on the conclusions that they generate."

Two other U.S. Supreme Court cases—*General Electric Company v. Joiner* (118 S.Ct 512 [1997]) and *Kumho Tire Company v. Carmichael* (119 S.Ct 1167 [1999])—build on the *Daubert* ruling to form a so-called "*Daubert* trilogy." These cases emphasize that the test of reliability is flexible and that the *Daubert* list of specific factors does not necessarily apply to all experts or in every case (see Harmon 2005).

The laboratory error rate issue raised in the *Daubert* decision's third factor is a difficult point to address (see Michaelis et al. 2008, p. 225). Proficiency tests, which test an individual analyst's ability to get the correct answer on a specific test using the laboratory's standard procedures, will not provide a reliable estimate of the probability of a laboratory error. Most courts consider the issue of possible laboratory error to influence the weight of the evidence rather than the admissibility (Michaelis et al. 2008, p. 227).

Charlotte Word comments: "Many types of errors can occur in a laboratory, but these are mostly human error. There are few errors in the technology with the exception of maybe stochastic effects and allele drop-in under low copy number conditions (see Chapter 11). Most laboratory standard operating procedures (SOPs) and quality control measures are designed to decrease the possibility of error and/or alert analysts to possible problems or errors. In the end, as recommended by the 1996 NRC II report, re-testing remains the best option if there is any reason to think an error was made. Thus, *it is best to save a portion of sample whenever possible for independent re-testing*. However, although the possibility of re-testing is an option, the burden is not on the defense to perform re-testing just because a sample remains. The burden is on the prosecution to "prove" their case in court" (CJW).

Summary of Admissibility Rulings

In summary, we have (1) *Frye* admissibility standards that emphasize general acceptance of theory and practice, (2) the Federal Rules of Evidence (FRE 702) that focuses on "helpful, relevant, and reliable" evidence, and (3) *Daubert* that makes the judge the gatekeeper for admitting evidence.

A study conducted 10 years after the June 28, 1993 *Daubert* ruling found a division in its application across the United States (Keierlber & Bohan 2005). This study classified the 50 states into (1) *Frye* states of which there were 15 (and 10 with codified evidence rules patterned after the FRE); (2) *Daubert* states of which there were 26 states (and 24 with FRE-based rules), and (3) non-*Frye*/non-*Daubert* states of which there were 9 (and 7 with FRE-based rules).

In commenting on what kinds of challenges are seen in admissibility hearings, Charlotte Word states: "In most situations, the challenges are the same seen routinely during trials, but with more emphasis on the unreliability of the technique or the limitations of the test system for forensic samples in general. The "newness" of the technique to the analyst and/or to the particular laboratory is often discussed with the focus being that something new must not be trustworthy and that the lack of extensive prior experience must lead to a higher risk of mistakes. As in trials, often hypothetical situations and "what if" questions are brought up. The specific challenges introduced will largely reflect the scientific background and knowledge of the expert testifying for opposing counsel. Reading any reports, affidavits, declarations, and/or prior testimony from the expert testifying for opposing counsel can be very helpful for understanding in advance what types of challenges to prepare for" (CJW).

TYPES OF DNA CASES AND OPPOSING COUNSEL TACTICS

Most forensic DNA scientists will be testifying for the prosecution particularly since the result of their DNA test can be a major or corroborating reason that a defendant is being prosecuted. If the DNA test result did not match a suspect, they would in most cases be exonerated and there would not be a trial. However, it has been noted that there are different types of cases where the DNA evidence might just be confirmatory or DNA may be the primary reason that a trial is being held at all. In addition, there will be varying degrees of defense challenges depending on the type of case and the defense lawyers and experts involved.

Categories of Cases Involving DNA

In explaining the approaches to presenting DNA evidence, one prosecutor has grouped the situations where DNA evidence is presented into three basic types of trials (Strange 2002):

> **Type #1**: The victim can identify the defendant or there is other equally compelling evidence establishing guilt. The DNA evidence is just confirming the other evidence.
>
> **Type #2**: The DNA evidence consists of a 9 or 13 [locus] STR match and is straight forward. The statistical power of discrimination in the match is trillion fold. However, without the DNA evidence, there is no case.
>
> **Type #3**: The DNA evidence may be a mixture (and there's little other evidence of guilt), mitochondrial DNA, "Y" chromosome, or SNP's. It's unlikely any of these will be used unless the DNA evidence is crucial.

He goes on to emphasize: "Quite clearly in Type 1 cases the DNA direct examination should consist of no more than the following: (a) Who are you and how employed? [name and job title, entity you are with], (b) What type of work do you do? [the very basics of your

job description], (c) How did you become qualified to do that? [education, training, and experience], (d) How do you do what you do? [Explain DNA typing and how you perform it in 5 minutes or less. Do not mention capillary electrophoresis by name.], (e) What did you do in this case? [explain what samples were tested and compared], and (f) What was the result? [same DNA profile, etc.]. The principle difference in the presentation of a Type 2 case is that more time needs to be spent on the credentials, particularly how often the examiner has performed this work" (Strange 2002).

These categories represent the opinion of a single prosecutor almost a decade ago and are probably too simplistic in today's world. However, the approach of categorizing cases can be helpful in spite of the fact that DNA evidence is often more complex now with improvements in technology and sensitivity.

Discovery Requests

As mentioned earlier in the chapter, discovery involves a request made (most often) by the defense counsel for copies of records used by the prosecution in their case against a defendant. Right to information against an accused individual goes back to the Sixth and Fourteenth Amendments to the U.S. Constitution (see earlier in chapter).

A 1963 Supreme Court case known as *Brady v. Maryland* (373 U.S. 83, 10 L.Ed.2d 215, 83 S.Ct. 1194 [1963]) mandates that prosecutors provide the defendant with copies of any exculpatory evidence in a timely manner (for some interesting background on the *Brady* case, see Klinkosum & Bannon 2006). The *Brady* ruling states that "the suppression by the prosecution of evidence favorable to an accused upon request violates due process where the evidence is material either to guilt or punishment." Michaelis and colleagues in their book *A Litigator's Guide to DNA* note that "if exculpatory evidence comes to the states' attention after conviction, the prosecutor has a responsibility to provide this evidence to the defense" (Michaelis et al. 2008, pp. 228-229).

In terms of DNA results, this request is typically for copies of the case report as well as the underlying data and case notes for all samples tested as part of the case—even those deemed to be inconclusive. Laboratory protocols and quality assurance measurements are often included in a discovery request. In many cases, information from validation studies or proficiency test results may also be requested. Discovery requests from the defense should be made through the prosecutor rather than directly to the laboratory. Although in some cases there may be pushback in terms of delivery of all requested material, the court expects responses to discovery requests to be complete and timely—that is delivery to the requesting party in a reasonable time period.

Prosecutors from the Orange County DA's DNA Unit remark: "Generally speaking, the defense is entitled to receive all discovery materials that would help them prepare their case. So, if the items requested are relevant, they should be provided. The law in California includes provisions that state that the defense must first approach the prosecutor for discovery materials. Upon occasion, discovery requests are improperly sent directly to the crime lab by the defense and are responded to by crime lab personnel without prosecutorial input. Without a thorough understanding of the law and a complete review of entire prosecution and investigation case files, a forensic scientist is not in the best position to make decisions regarding the release of discovery materials" (OCDA).

While turning over your results and underlying validation studies may seem disconcerting, discovery should not be feared. It is part of the due process of law guaranteed under the Fourteenth Amendment of the U.S. Constitution and emphasized in the 1963 U.S. Supreme Court case *Brady v. Maryland*. If you are confident in your work product and protocols as a DNA analyst, sharing your data should not be a problem. As noted by the NRC II report, "there are no strictly scientific justifications for withholding information in the discovery process" (NRC II 1996, p. 167).

Robin Cotton shares: "I think the most important question for a lab (in fact, any discipline in the lab), is if you were to hand this information—the information you are going to give over to the defense—to a respected colleague, would you have given them everything they need to evaluate your work? If the answer is "yes", then that's about what you should be giving on discovery. For DNA that might be your case folder, your procedures, and your electronic data. I know that the defense often asks for lots of other documents: validation studies, proficiency test results, etc. Different states have different rules and different prosecutor's offices have different attitudes about giving over that extra material…. In any case, you should give over, *without question and happily*, the things that you would need from another competent DNA person to perform the review. *Do not worry about who might look at the information as a defense expert!*" (RWC).

With the advent of computer collection of DNA typing data (see Chapter 6), electronic data files, rather than just copies of DNA results, are requested by some defense experts for the purpose of reanalyzing the data to see if the same conclusions can be drawn. Charlotte Word notes: "Electronic data should be turned over if the profiles are not present in the case file or if they are not in a format that an independent expert can sufficiently review the data obtained to assess how the laboratory came to the decisions made" (CJW).

Sample Retesting

If any untested sample remains after the initial evidence examination, the defense has a right to go through the court to get access to the evidence for re-testing and perform its own independent testing provided sufficient sample is available to do so. The NRC II committee in its 1996 report stated, "A wrongly accused person's best insurance against the possibility of being falsely incriminated is the opportunity to have the testing repeated. Such an opportunity should be provided whenever possible" (NRC II 1996, p. 87). NRC II recommendation 3.3 states: "Whenever feasible, forensic samples should be divided into two or more parts at the earliest practicable stage and the unused parts retained to permit additional tests. The used and saved portions should be stored and handled separately. Any additional tests should be performed independently of the first by personnel not involved in the first test and preferably in a different laboratory" (NRC II 1996, p. 88). The committee concludes that: "A defendant who believes that the match is spurious should welcome the opportunity for an independent repeat test" (NRC II 1996, p. 87).

Charlotte Word summarizes: "Although the possibility of re-testing is an option for the defendant, the burden of proof is on the state to "prove" their case beyond a reasonable doubt. The defense attorney is not required to prove the defendant's innocence since that is already presumed for the defendant under the U.S. Constitution. Therefore, the defense attorney has no responsibility or obligation to perform re-testing just because a sample

remains. In many jurisdictions and cases, if the defense attorney requests re-testing and decides to call the expert who tested the evidence as a witness at trial, those results must also be provided to the prosecution. The attorney would therefore not be acting in the best interest of his/her client to request re-testing since it cannot be known prior to re-testing that different results would be obtained" (CJW).

Common Opposing Counsel Tactics

Table 18.4 lists several common lines of attack on forensic DNA testing results and testimony offered in court including chain-of-custody, qualifications of the expert, reliability of the technique for testing evidence, reliability of the analyst(s) performing the work, and reliability of the laboratory in which testing was performed including the standard operating procedures developed therein. Potential contamination of evidence and/or test results may be raised as a concern. The statistical calculations performed may also be questioned including whether or not appropriate equations or data were used or appropriate population(s) allele frequencies for the perpetrator versus the defendant.

Some other common challenges to DNA results and testimony include mishandling of evidence, human error during testing, bias in testing (getting "the answer" that is wanted by the client requesting the testing), lack of information regarding when and how the DNA was deposited on an item, and lack of control over the handling of the evidence prior to receipt in the laboratory.

Ted Hunt notes: "Defense challenges come in a variety of forms depending on the nature of the case and the significance of the DNA evidence. If the DNA results are clean and fairly unassailable, a popular defense tactic is to concede that the detected profile belongs to the defendant but that its presence at the scene was the innocuous result of a secondary or tertiary transfer… When the evidentiary context of the DNA leaves little room for innocuous explanations, the attacks turn to evidence handling, the testing process, and the interpretation of the results…" (TRH).

This seasoned prosecutor concludes: "By and large I would say that these tactics are ultimately not successful. I have yet to lose a case involving DNA evidence. Juries may have

TABLE 18.4 Common Challenges to DNA Results and Testimony.

- Chain-of-custody
- Qualifications of the expert
- Reliability of the technique for testing evidence
- Reliability of the analyst(s) performing work
- Reliability of the laboratory in which testing was performed, SOPs
- Contamination of evidence and/or test results
- Mishandling of evidence
- Human error during testing
- Bias in testing (getting "the answer" that is wanted by the client requesting testing)
- Statistical calculations, appropriate population(s) for the perpetrator vs. defendant, appropriate equations used, appropriate data used
- Lack of information regarding when and how the DNA was deposited on an item, and lack of control over the handling of the evidence prior to receipt in the laboratory

more information to sort through at the end of a case when a defense DNA expert is called, but a well-prepared state's expert will usually win the day in court" (TRH).

Brad Bannon offers his perspective as a defense attorney: "When the evidence supports challenging a DNA expert's conclusion, you must demystify the process and demonstrate to the jury that DNA analysis is a subjective human exercise. Contamination and chain-of-custody arguments are only the tip of the iceberg. It's important to portray the entire process for what it is: one that involves human beings at all levels, from sample handling, to sample processing, to sample interpretation, to final reporting. Each level is vulnerable to human error or bias, and so is the analyst's courtroom testimony" (BB).

He continues: "If there were steps along the way when the analyst deviated from scientific standards, lab protocols, or the analyst's own customary practice, or if the analyst made an interpretation or judgment call that could have gone either way but ended up favoring the prosecution, the lawyer should be prepared to highlight the importance of those things for the jury and, if necessary, confront the analyst about them in the courtroom. While it's never my intent to get into a "contest" with a DNA expert about who knows more about DNA, my goal is to have a meaningful dialogue with the expert about the case-specific DNA issues and to make the points I set out to make—or even a few new ones, should the opportunity present itself—without being intimidated by the knowledge gap" (BB).

Forensic scientists facing defense attorneys in court may become frustrated during cross-examination and not fully understand the important role they play in the criminal justice system. More than three decades ago, some wise advice was published in the *Journal of Forensic Sciences*: "The forensic scientist should see the defense lawyer not as his enemy but as an officer of the court sworn to do the best job possible for his client, regardless of whether the client has retained the lawyer or whether the defense lawyer has been appointed by the court or is a public defender" (Phillips 1977).

As the next section describes, one important lesson we can learn from the recent post-conviction exonerations enabled by DNA testing is that not everyone who was prosecuted and convicted in the past was "guilty." It is hoped that as we go forward with better science in the future, this tragedy of justice can be avoided. The expert witness has an important role to be vigilant in efforts to support the data accurately, without overstatement, and not be biased towards either the prosecution or the defense.

Post-Conviction Testing

DNA testing has played an important role in post-conviction exonerations. Ted Hunt states: "To the extent that DNA evidence has been able to help exonerate the truly innocent, everyone has reason to celebrate" (TRH). Brad Bannon adds: "I'm not sure anything has had a greater impact on the criminal justice system since I've been practicing law. Though skepticism of criminal accusation has long been embodied in the presumption of innocence and the requirement of proof beyond a reasonable doubt for conviction, collecting all of the most eloquent speeches in the world about *why that skepticism is right* does not add up to the power of one, let alone dozens, let alone hundreds, of stories about people who served 5, 10, 15, sometimes up to 30 years for crimes they did not commit" (BB).

Brad Bannon continues: "Because many of the original convictions in the DNA exoneration cases were based on mistaken eyewitness identification, it has also given stakeholders in the

criminal justice system—judges, prosecutors, defense attorneys, law enforcement—a healthy concern about such evidence and has led to reforms in many eyewitness identification procedures across the country. As importantly, in addition to exonerating the innocent, post-conviction DNA analysis has also resulted in the identification, apprehension, and conviction of the actual perpetrators of the crimes. After all, forensic science is not just about making sure that the wrong guy does not get convicted; it's about making sure the right one does" (BB).

It is important for the integrity of evidence to be maintained for it to be useful in these old cases. Old samples, which were frequently handled without gloves in the laboratory or by attorneys, witnesses, etc., in court, can lead to contamination. In these old cases, elimination samples from legitimate people who may have come in contact with these samples may not be possible to obtain. Issues can arise if the DNA results exclude the convicted defendant. Is the post-conviction DNA evidence result from the profile of the true perpetrator, thus proving the innocence of the individual in prison? Or can the DNA results be explained by other means such as another individual with access to evidence but not associated with the crime (e.g., consensual partner, contamination during evidence handling, testing, or storage)?

Contamination might only rarely impact a careful forensic DNA laboratory. However, it can have potential significance in old cases under review including those that are part of the Innocence Project. For example, if biological evidence from a 20-year-old case was handled by ungloved police officers or evidence custodians (prior to knowledge regarding the sensitivity of modern DNA testing), then the true perpetrator's DNA might be masked by contamination from the collecting officer. Thus, when a DNA test is performed, the police officer's or evidence custodian's DNA would be detected rather than the true perpetrator. In the absence of other evidence, the individual in prison might then be falsely declared "innocent" because his DNA profile was not found on the original crime scene evidence. This scenario emphasizes the importance of considering DNA evidence as an investigative tool within the context of a case rather than the sole absolute proof of guilt or innocence.

JURY COMPREHENSION OF DNA EVIDENCE

Over 30 years ago, a serologist from Palm Beach Sheriff's Crime Laboratory wrote (Tanton 1979): "When the forensic scientist testifies in court, both he and his testimony are usually evaluated by a group of people that has never seen him before and has little or no scientific background. These people, the jury, are exposed to the forensic scientist for only a relatively short time, during which they are privy to a strictly regulated question-and-answer conversation carried on between the scientist and the attorneys in the case. As a result of this exposure, the jurors must answer several important questions:

1. What did the expert say?
2. What is the significance of the expert's testimony?
3. Is the expert competent?
4. Is the expert honest?

Once they have answered these questions, and they always do, either directly or indirectly, they must decide what weight to give his evidence in their total deliberation" (Tanton 1979).

Testimony from forensic DNA analysts can be overwhelming to jury members—who often have limited exposure to the concepts of biology, genetics, and the technology used to generate DNA profiles. Lisa Kahn and David Feldstern from the Los Angeles County District Attorney's Office note that "When DNA testimony is properly presented, most jurors feel empowered by their newly found understanding of this important forensic evidence" (Kahn & Feldstern 1998).

When asked how you know whether or not you are connecting with the jury, Robin Cotton responded: "You can just forget it. You don't know" (RWC). Remember that it takes effort to select appropriate, understandable words to describe the DNA testing process in lay terms rather than scientific jargon. Take time to carefully consider the best words to use to describe your results. How you communicate will benefit juror comprehension.

The National Research Council's 1996 publication entitled *The Evaluation of Forensic DNA Evidence* (NRC II) recommended that research be conducted on juror comprehension of DNA evidence. NRC II recommendation 6.1 states: "Behavioral research should be carried out to identify any conditions that might cause a trier-of-fact [judge or jury] to misinterpret evidence on DNA profiling and to assess how well various ways of presenting expert testimony on DNA can reduce any such misunderstandings" (NRC 1996).

A 2004 study in Australia found that juries were 23 times more likely to vote guilty in homicide cases and 33 times more likely to vote guilty in sexual assault cases when DNA evidence was admitted in the trial (Briody 2004). Several such studies have been performed including one aimed at seeing if jurors comprehended that a mitochondrial DNA (mtDNA) match had less strength than a nuclear DNA result (Dann et al. 2004).

Expert witnesses need to connect with the jury and teach them at a level that they can comprehend the information shared. Results should be confidently conveyed but not overstated. New Zealand scientists John Buckleton and James Curran have written: "There is a considerable aura to DNA evidence. Because of this aura it is vital that weak evidence is correctly represented as weak or not presented at all" (Buckleton & Curran 2008). Where there is uncertainty in the data (e.g., with complex DNA mixtures, partial profiles from degradation, or low level DNA results), this should be emphasized so that the jurors (and attorneys for both sides) may appropriately evaluate the weight of the evidence presented.

As recommended in SWGDAM Interpretation Guideline 4.1 (SWGDAM 2010): "The laboratory must perform statistical analysis in support of any inclusion that is determined to be relevant in the context of a case, irrespective of the number of alleles detected and the quantitative value of the statistical analysis." Clearly a "1 in 300" statistic does not have the same weight as "1 in 300 billion." Make sure that the court—and especially the jury who will be making a decision based on your information and other aspects of the case presented to them—understands what your data says and what it does not say.

FACING THE "CSI EFFECT"

The so-called "CSI effect" is prevalent in juries today because of the popularity of crime solving TV shows (Houck 2006, Durnal 2010, Holmgrem & Fordham 2011). CBS's "CSI: Crime Scene Investigation" has been a top-rated show on television for many years. CSI

and other crime shows have made forensic science seem easy and rapid. These television dramas influence the general public to a point where they expect all cases to be solved with forensic evidence—and especially DNA results—when they serve on a jury. This false expectation can make life challenging for prosecutors trying to prove their cases. Equally important is that jurors may have a false perception of what forensic scientists outside of Hollywood can and cannot accomplish.

FINAL THOUGHTS

When I asked Dr. Robin Cotton what was the single most important thing to remember when serving as an expert witness, she replied: ***"You are the voice of the data!*** You are not a voice for the victim, which is what some prosecutor's describe their role as. People will talk about the 'prosecution team.' I really don't think you are on somebody's team, but that isn't to say you wouldn't assist them in understanding or assist them in answering questions so that the data is clear to the jury. Just because you help in appropriate ways, it doesn't mean you are on 'their side.' If you want to feel like you are on someone's side, then you are welcome to feel so. You just cannot behave like you are on their side. You cannot let that feeling influence how you behave, how you speak, and most importantly, how you look at the data. I think it is a hard issue because you want to please the people you are working with, right? It is the prosecutor who is supposed to worry about the consequences of the trial. If you represent the data accurately in a scientific sense, then it is hard to go wrong" (RWC).

In their *Profiles in DNA* article, Lisa Kahn and David Feldstern conclude: "In many ways, the expert witness has the easiest job in the courtroom. If you are qualified and prepared, testifying should be an enjoyable and rewarding experience. An outstanding expert witness is one whose demeanor does not change from direct to cross-examination. If a person walks into the courtroom while you are testifying, they should find it difficult to tell whether you are still under direct examination, or are under cross-examination" (Kahn & Feldstern 1998).

Good luck with your experience in court as an expert witness! I hope that the material provided in this chapter has been helpful. Please see Appendix 4 for additional information from those interviewed in preparing this chapter.

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